Certified Engine-A Addendum for RQ-000013

to the supplementary material for
Effective Archimedean Stark Theorems over Real Quadratic Fields

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31 July 2026

Claim boundary

The identity below is `PROVED`: the uniform Engine-A theorem is applied after every case-specific hypothesis is checked by exact arithmetic. The accompanying `bnrL1` point-value comparison is only an `OBSERVED` cross-check and is not used in the proof. This versioned addendum stages the row for the next authorized release; it does not alter the already-published Zenodo v1.3 files.

The first nonzero imprimitive branch

Let $K = \mathbb{Q}(\sqrt{2})$, with finite ideal in the frozen integral basis

$$\mathfrak{f} = \begin{pmatrix} 14 & 6 \\ 0 & 2 \end{pmatrix}, \qquad N\mathfrak{f} = 28,$$

and modulus $\mathfrak{m} = \mathfrak{f}\infty_2$. Its ray group is $G \simeq C_2$, in pinned PARI coordinates $G = \{[0], [1]\}$; the sign class is $R = [1]$. The unique supported character has $\chi([1]) = -1$. Its primitive conductor is

$$\mathfrak{f}_\chi = \begin{pmatrix} 7 & 3 \\ 0 & 1 \end{pmatrix} \infty_2.$$

The unique omitted prime has norm 2, valuation 2 in $\mathfrak{f}$, valuation 0 in $\mathfrak{f}_\chi$, and $\chi^\circ(\mathfrak{p}) = -1$. Hence the exact imprimitive factor is

$$E_\chi = 1 - \chi^\circ(\mathfrak{p}) = 2.$$

The quadratic ray field is represented relatively and absolutely by

$$t^2 - 2\sqrt{2} + 1, \qquad P(t) = t^4 + 2t^2 - 7.$$

It has signature $(2, 1)$, discriminant -448, class number 1, and two roots of unity; `bnfcertify` returns 1. On its free-unit basis the exact norm map is

$$\begin{pmatrix} 1 & -1 \end{pmatrix},$$

whose primitive kernel is generated by $(1, 1)^T$. The embedded base unit has coordinates $(1, -1)^T$, so

$$I_\chi = \left| \det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \right| = 2.$$

Put

$$u = \frac{t^2 - 2t + 3}{4}.$$

The chosen absolute generator is the unique root of P in $(-34/25, -27/20)$. Exact Sturm counts and rational endpoint evaluation give

$$\frac{9}{5} < u < \frac{19}{10}.$$

Thus this is the oriented relative unit. It has minimal polynomial

$$U^4 - 2U^3 + U^2 - 2U + 1,$$

and the nontrivial Artin element $t \mapsto -t$ sends u exactly to u^{-1}.

Since $h_K = h_L = 1$, $w_K = w_L = 2$, $E_\chi = 2$, and $I_\chi = 2$, the uniform theorem gives

$$L'_{\mathfrak{m}}(0, \chi) = 2 \log u.$$

Here $|G| = 2$, so the Fourier coefficient $2/|G|$ is one. Therefore the fully Artin-labelled packet is

$$\boxed{X_{[0]} = u^2, \qquad X_{[1]} = u^{-2}.}$$

The selected value $u^2 \in (7/2, 18/5)$ has minimal polynomial

$$X^4 - 2X^3 - 5X^2 - 2X + 1.$$

Replay record

The source of truth is the JSON record whose repository-relative path is split here only for typesetting:

`artifacts/rq000013-engine-a-`
`imprimitive-certificate-v1.json`

Its deterministic replay command is

```
python3 scripts/certify_rq000013_engine_a.py --check
```

under PARI/GP 2.15.4. The frozen transcript is at

`artifacts/rq000013-engine-a-`
`imprimitive-certificate-v1.transcript.`